

THE THREE CANONICAL MACHINE ARCHETYPES OF PHYSICAL AI

Distilling Physical Systems by Governing Physics, Bounding Budgets, and Primary Failure Modes

1. MASS-DOMINATED

Mobility & Navigation



$$d_{stop} = v_0 \Delta t + v_0^2 / (2 a_{max})$$

Exemplars: AMRs, AVs, Drones, Cranes

First Budget: Stopping Distance & Freshness

Failure Mode: Kinematic boundary crash

Safety Gate: Real-time dynamic braking referee

Delay converts directly to spatial drift (Δx)

2. CONTACT-DOMINATED

Manipulation & Assembly



$$F_c \approx k_c \Delta x$$
$$J_{ref} = N^2 J_r$$

$$F_{contact} \approx k_c \Delta x, J_{ref} = N^2 J_{rotor}$$

Exemplars: Manipulators, Grippers, Humanoids

First Budget: Actuator Bandwidth & Gear Stress

Failure Mode: Gear tooth shear, workpiece damage

Safety Gate: Torque clamping & C² jerk profiler

Position errors convert to massive forces (k_c)

3. FLOW-DOMINATED

Process, Thermal & Energy



$$D \leq (T_{max} - T_{amb}) / (I^2 R \theta_{JA})$$

Exemplars: Thermal loops, Inverters, Bioreactors

First Budget: Thermal Capacity & Phase Timing

Failure Mode: Thermal runaway, insulation blowout

Safety Gate: Hardware thermal observer & interlock

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